Multi-sensor Fusion SLAM Algorithm Considering Wheel Slip

Song Xiaobo
School of Automation
Wuhan University of Technology
Wuhan, China
15671664301@139.com

Chen Ziming
School of Automation
Wuhan University of Technology
Wuhan, China
2597877213@qq.com

Du Luyao
School of Automation
Wuhan University of Technology
Wuhan, China
1586886748@qq.com

Xiang Kui
School of Automation
Wuhan University of Technology
Wuhan, China
xkarcher@126.com

2024 30th International Conference on Mechatronics and Machine Vision in Practice (M2VIP) | 979-8-3503-9191-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/M2VIP62491.2024.10746072

***Abstract*—With the continuous development of automated guided vehicle (AGV) technology, simultaneous localization and mapping (SLAM) has become a research hotspot. Currently, domestic AGVs mainly construct environmental maps based on laser radar scanning, while inertial measurement unit (IMU) sensors and wheel odometry are used to provide attitude estimation and motion compensation, speed and displacement information respectively. However, when the drive wheels of the AGV slip, the wheel odometry can produce significant errors, leading to positioning errors and map drift. This paper uses an error state Kalman filter (ESKF) to fuse wheel-IMU odometry and a slip detection method based on slip rate. When the AGV slips, the wheel odometry data is promptly discarded, and the IMU pre-integration is used directly to replace the odometry, effectively reducing the impact of slipping. Through comparative experiments, it can be found that the algorithm can reduce the error caused by slip by about 78%, which has certain stability and accuracy, and can further improve the robustness of the system when applied to AGV.**

Keywords—SLAM; ESKF; AGV slip detection; Laser odometry

I. INTRODUCTION

The research and application of AGV system technology involve multiple fields, with the characteristics of complexity and universality. AGVs are high-tech products that integrate mechanical, electronic, and computer technologies. The key technologies of AGVs include motion control technology, navigation technology, and upper-level control technology. Motion control technology is used to achieve precise movement and operation of AGVs, while navigation technology determines the position and direction of AGVs in the work environment. Upper-level control technology encompasses core areas such as task scheduling, vehicle allocation, path planning, and traffic management. These technologies ensure that AGVs can efficiently perform various tasks and coordinate and interact with other AGVs or systems.

With the continuous progress of technology, the application fields of AGV are constantly expanding, and various technologies are also constantly upgrading, which makes the functions of AGVs more diverse and the performance is constantly improving. AGVs have strong functionality and practicality, capable of handling heavy loads ranging from tens of kilograms to hundreds of tons. According to the needs of different working environments, they can be classified into different types such as general indoor, low-temperature environments, and outdoor environments. There are also multiple choices in load transfer methods [1], driving methods, and power supply methods [2] to adapt to different needs and environments. In terms of navigation, the AGV system adopts various technologies, including magnetic tape navigation, laser navigation [3], inertial navigation [4], and visual navigation [5][6]. These technologies each have their own advantages and can be selected and applied according to specific circumstances.

In terms of navigation, AGV systems employ various technologies, including magnetic tape navigation, laser navigation, inertial navigation, and vision navigation. Each of these technologies has its advantages and can be selected and applied based on specific situations. Laser radar obtains precise distance information from the environment by emitting laser beams and receiving the reflected light, thereby constructing detailed maps. However, laser radar can be affected by environmental factors such as lighting and obstructions, which can reduce positioning accuracy in specific situations. Inertial navigation systems estimate the attitude and displacement of the vehicle by measuring its acceleration and angular velocity, offering high real-time performance. However, inertial navigation systems have the issue of integration errors, where errors accumulate over time, affecting positioning accuracy. Wheel odometry can provide speed and displacement information for AGVs, but significant errors occur when AGVs slip.

In this paper, in order to fully utilize the advantages of various sensors and improve the overall performance of AGV positioning and mapping, the multi-sensor fusion SLAM algorithm is studied. By integrating the information from various sensors at the algorithmic level, the deficiencies of individual technologies can be compensated, enhancing the robustness, accuracy, and real-time performance of the system.

II. ONBOARD SENSOR MODEL

A. Lidar model

This paper uses the WLR-720 laser radar, which employs the Time of Flight (TOF) principle to obtain detailed information about the surrounding environment. The TOF working principle involves the laser emitter sending out laser beams to the surroundings. When the laser beams hit the surface of an object, a reflected light signal is generated, which is then received by a specific sensor. By measuring the time interval between the emission and reception of the laser, and multiplying it by the speed of light, the distance to the target object can be accurately calculated [7]. The example diagram of laser ranging based on TOF principle is shown in Figure 1. The laser emitter is responsible for sending out laser beams, the receiver collects the laser beams reflected from the object, and the timer provides the time difference between the emission and reception of the laser signal. Using this information, the position of the point cloud in the laser radar coordinate system can be accurately calculated.

As shown in Figure 2,The three-dimensional spatial coordinates of point P can be calculated using the following formula:

$$X = Rcos(\omega)sin(\alpha) \quad (1)$$

$$Y = Rcos(\omega)cos(\alpha) \quad (2)$$

$$Z = Rsin(\omega) \quad (3)$$

The R in the formula is the maximum detection range of the LiDAR, ω is the scanning angle of the LiDAR in the vertical direction, and α is the scanning angle of the LiDAR in the horizontal direction.

B. IMU model

The IMU module selected in this paper is the WLR-720 with a built-in IMU chip, model ASM330LHH, which is a 6-axis automotive-grade IMU capable of outputting angular velocity and acceleration data. The IMU data output frequency is 100Hz. This paper uses IMU data output driven by ROS, with completed calibration for temperature drift and zero drift, ensuring data stability and accuracy. The IMU data output coordinate system is consistent with the coordinate system of the radar point cloud data, as shown in Figure 3.

The measurement model of the accelerometer [8] is:

$$\hat{\alpha}_k = \alpha_k + b_{\alpha k} + R_W^{B_k} g^W + \eta_a \quad (4)$$

In the formula, $\hat{\alpha}_k$ represents the accelerometer measurement value in the IMU coordinate system of the k-th frame, α_k represents the theoretical acceleration value in the k -th frame, α_k represents the zero point error of the

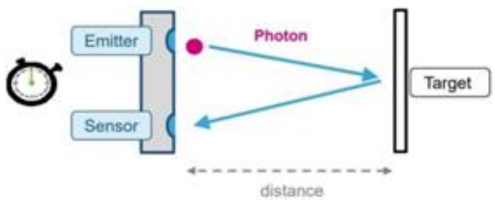

Fig. 1. TOF laser ranging schematic diagram

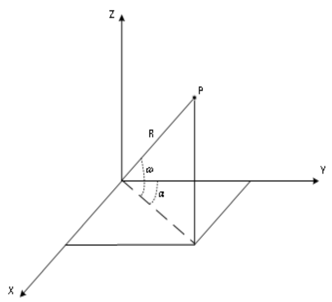

Fig. 2. Lidar sensor coordinate system

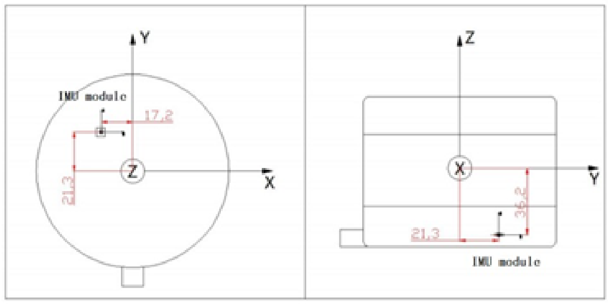

Fig. 3. Lidar sensor coordinate system

acceleration that needs to be estimated online in the k-th frame, $b_{\alpha k}$ represents the system (body) coordinate system, which is the IMU coordinate system, W represents the world (world) coordinate system, $R_W^{B_k}$ represents the rotation matrix of the transformation from the world coordinate system to the body coordinate system,$g^W = [0\ 0\ g]^T$ represents the gravitational acceleration in the world coordinate system, g represents the local gravitational acceleration, and η_a represents the acceleration white noise.

The measurement model of the gyroscope [9] is:

$$\hat{\omega}_k = \omega_k + b_{gk} + \eta_g \quad (5)$$

In the formula, $\hat{\omega}_k$represents the gyroscope measurement value at frame k, $\hat{\omega}_k$represents the theoretical angular velocity at frame k, b_{gk} represents the zero point error of angular velocity at frame k that needs to be estimated online, and η_g represents the white noise of angular velocity.

C. Wheel odometry model

The encoder is a critical sensor designed to collect information such as wheel speed, motor temperature, bus voltage, and current. By utilizing the speed of each wheel and applying the kinematic model of the AGV, it is possible to determine the AGV's body speed and odometry. The encoders used in this paper are integrated into the AGV's drive module, with a sampling frequency of 200Hz. They communicate data to the upper-level computer via the CAN bus.

Based on the kinematic model of wheel speed, wheel radius, and AGV, the distance of the wheel relative to the ground and the total distance traveled by the vehicle can be obtained. The dual wheel differential kinematic model is a simple and universal kinematic model, as shown in Figure 4. In this model, the X, Y, and Z axes represent the front, right, and upper directions of the AGV body, respectively. The origin O is located at the center of the wheel axle, l is the track width, θ is the yaw angle, and v_1 and v_2 represent the linear velocities of the left and right wheels, respectively.

Based on the wheel speed, wheel radius, and the kinematic model of the AGV, the distance that the wheels move relative to the ground can be determined, and the total distance traveled by the vehicle can also be calculated [10]. The differential drive kinematic model is a simple and widely used model, as shown in Figure 4. In this model, the X, Y, Z axes represent the forward, right, and upward directions of the AGV chassis respectively, with the origin O located at the center of the axle. l is the wheelbase, θ is the yaw angle, and v_L, v_R represent the linear velocities of the left and right wheels, respectively.

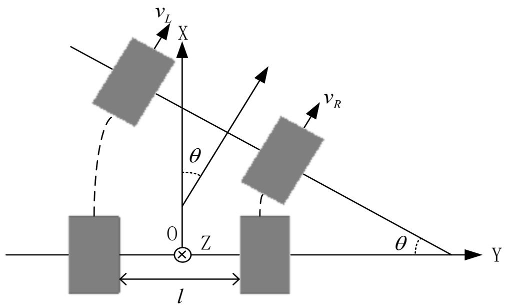

Fig. 4. Dual wheel differential kinematic model

Assuming that the radii of the left and right wheels are r_L and r_R respectively, and the angular velocities ω_L and ω_R respectively:

$$v_L = r_L \omega_L \tag{6}$$

$$v_R = r_R \omega_R \tag{7}$$

The speed v and yaw angle ω of AGV are respectively:

$$v = \frac{v_L + v_R}{2} \tag{8}$$

$$\omega = \frac{v_L - v_R}{l} \tag{9}$$

III. WHEEL SPEED IMU ODOMETRY

In order to adapt to the characteristics of rapid movement, complex environment, and long-term operation of AGVs in industry, this paper uses ESKF technology. By effectively combining IMU data with wheel speed odometry data, the accuracy of positioning and the robustness of mapping can be effectively improved. One of the characteristics of the error state Kalman filter is that it processes relatively small error states, so the overall computational complexity is also relatively low, which has a small impact on the overall performance of the algorithm.

A. Sports state prediction

Based on the derivative relationship in the predicted state kinematic equation, the position, velocity, attitude, gyroscope deviation, and acceleration deviation of the IMU from time t to time K can be calculated by integrating the measured values of the IMU. In real measurements, continuous kinematic equations can be transformed into discrete forms through numerical integration, as shown in formulas (10) - (14):

$$p \leftarrow p + v\Delta t + \frac{1}{2}[R(a_m - a_b) + g]\Delta t^2 \tag{10}$$

$$v \leftarrow v + [R(a_m - a_b) + g]\Delta t \tag{11}$$

$$q \leftarrow q \otimes q[(\omega_m - \omega_b)\Delta t] \tag{12}$$

$$a_b \leftarrow a_b \tag{13}$$

$$\omega_b \leftarrow \omega_b \tag{14}$$

Convert the kinematic equation of the error state into discrete form, as shown in formulas (15) - (19):

$$\delta p \leftarrow \delta p + \delta v \Delta t \tag{15}$$

$$\delta v \leftarrow \delta v + (-R[a_m - a_b] \times \delta\theta - R\delta a_b + \delta g)\Delta t + v_i \tag{16}$$

$$\delta\theta \leftarrow R^T[(\omega_m - \omega_b)\Delta t]\delta\theta - \delta\omega_b \Delta t + \theta_i \tag{17}$$

$$\delta a_b \leftarrow \delta a_b + a_i \tag{18}$$

$$\delta\omega_b \leftarrow \delta\omega_b + \omega_i \tag{19}$$

Therefore, it can be concluded that the error state propagation system:

$$\delta x \leftarrow f(x, \delta x, u_m, i) = F(x, u_m) \cdot \delta x + F_i \cdot i \tag{20}$$

The vector x represents the predicted state, $\delta_x$is the error state vector, u_m is the input vector, and i is the random disturbance pulse vector.

In error state Kalman filtering, the propagation equation of the error covariance matrix is obtained by linearizing the kinematics of the error state:

$$p \leftarrow F_x p F_x^T + F_i Q_i F_i^T \tag{21}$$

Among them, F_x and F_i are Jacobian matrices of f with respect to the error state δ_x and disturbance vector i , $F_x = \frac{\partial f}{\partial \delta x}$,$F_i = \frac{\partial f}{\partial i}$,and Q_i is the covariance matrix of the disturbance vector.

B. Motion status update

System observation model:

$$y = h(x_t) + v \tag{22}$$

Where $h()$ represents the nonlinear function of the system state, v represents Gaussian white noise, and $v \sim N(0, V)$.

Calculate the Kalman gain, update the total error state and error state covariance matrix, as shown in formulas (23) - (25):

$$K = PH^T(HPH^T + V)^{-1} \tag{23}$$

$$\delta\hat{x} \leftarrow K(y - h(\hat{x}_t)) \tag{24}$$

$$P \leftarrow (I - KH)P \tag{25}$$

Where $H = \frac{\partial h}{\partial \delta x}$.

Update the prediction status, as shown in formulas (26) - (30):

$$P \leftarrow P + \delta\hat{P} \tag{26}$$

$$v \leftarrow v + \delta\hat{v} \tag{27}$$

$$q \leftarrow q \otimes q(\delta\hat{\theta}) \tag{28}$$

$$a_b \leftarrow a_b + \delta\widehat{a_b} \tag{29}$$

$$\omega_b \leftarrow \omega_b + \delta\widehat{\omega_b} \tag{30}$$

Among them, P represents the position after state update, v represents velocity, q represents attitude, a_b and ω_b represent acceleration bias and angular velocity bias.

C. ESKF data fusion

The state dimension of the filter is 15, which includes biases for position, velocity, attitude, acceleration, and angular velocity. The advantage of ESKF is that it can effectively handle small quantities, which helps simplify computational complexity. In the fusion process, the system considers both the wheel speed odometry and IMU data, dynamically estimates the error state through iteration, and corrects it to the system state. The process is as follows:

1) Using the observation data of IMU as prior information, predict the covariance matrix of the nominal state, error state, and error state.

2) Using the information from the wheel speed odometry as observation information, calculate the Kalman gain matrix, and update the state and error state covariance matrix of ESKF.

3) Correct the predicted state with the predicted error state to obtain the position, attitude, velocity, acceleration deviation, and angular velocity deviation of six degrees of freedom

As shown in Figure 5, the data fusion process in ESKF consists of three steps: prediction, correction, and update. The data for the prediction process is provided by the IMU, while the data for the correction process is provided by the wheel speed odometry. The corrected error will be updated to the nominal variable. After the update is completed, the error state will be set to zero, completing multi-sensor data fusion.

According to the formula of ESKF, the observation equation can be derived to correct the k-th frame, and the Kalman gain can be calculated:

$$\delta\hat{x}^k = K^k(Z^k - H\Delta x^k) \tag{32}$$

Calculate the posterior value of the error state based on the calculated Kalman gain:

$$\delta\hat{x}^k = K^k(Z^k - H\Delta x^k) \tag{32}$$

And update the covariance P to pass it on to the next data fusion:

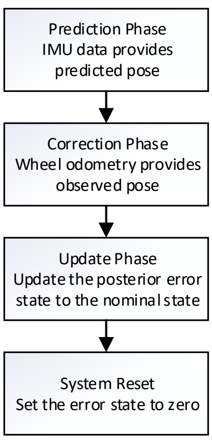

Fig. 5. ESKF process diagram

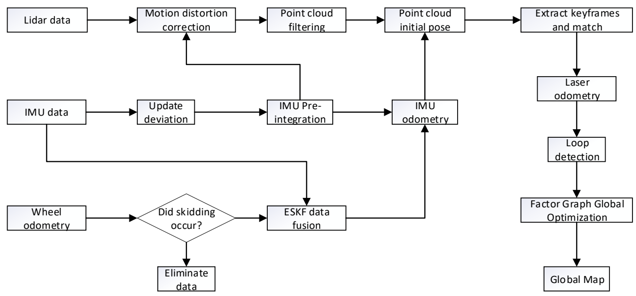

Fig. 6. Algorithm framework diagramm

$$P^k = (I - K^k H)P^k \tag{33}$$

After calculating the posterior error state value during the correction stage, update it to the nominal state x^k at that time to obtain the posterior nominal state $\hat{x}^k$:

$$\hat{x}^k = x^k \oplus \delta\hat{x}^k \tag{34}$$

After completing the update of the nominal state, it is considered that the current round of ESKF data fusion is completed, and the system needs to be reset, that is, the error state is set to zero, ready for the next prediction and correction:

$$\delta x = 0 \tag{35}$$

IV. MULTI SENSOR FUSION SLAM ALGORITHM

A. The overall process of the algorithm

The algorithm flowchart of this article is shown in Figure 6. When obtaining a frame of point cloud data, first perform motion distortion correction for point cloud preprocessing, then extract point cloud feature points, combine with the IMU odometry optimized in the previous chapter, perform loop detection and global factor graph optimization, and finally obtain a high-precision 3D point cloud map.

B. AGV slip detection based on slip rate

The measurement of the wheel speed odometry is based on the rotation of the wheels. If the wheel slips, the measured distance will not match the actual distance traveled. This may lead to errors in accumulating distance and estimating AGV position. Once slippage occurs, the established map may experience distortion or displacement, which may result in map discontinuity or loop detection failure when reflected on the global map.

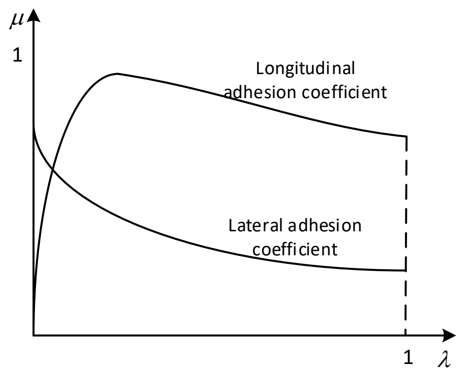

Fig. 7. Relationship between slip rate and adhesion coefficient

The slip rate of AGV is one of the important indicators for evaluating the stability and safety of vehicle operation. It reflects the adhesion between the tires and the ground during AGV driving, as shown in formula (36):

$$\lambda = \frac{\omega r_\omega - V}{\omega r_\omega} \tag{36}$$

In the formula, ω is the angular velocity of the AGV wheels, r_ω is the rolling radius of the wheels, and V is the longitudinal velocity of the vehicle body. According to the formula, the range of slip rate values is 0-1.

To solve the problem of AGV slippage, slippage detection is added to the SLAM system. The relationship between slip rate and ground adhesion coefficient is shown in Figure 7. The longitudinal adhesion coefficient increases and then decreases with the slip ratio, while the lateral adhesion coefficient decreases as the slip ratio increases. Numerous studies have shown that when the slip rate is between 0-0.2, it indicates stable operation of the AGV, while when the slip rate is greater than 0.2, it indicates severe slip of the AGV. At this point, it is necessary to eliminate the fusion of wheel speed odometry and IMU data, ignore the measurement from the wheel speed odometry, and directly use the original data from the IMU to avoid errors caused by wheel speed odometry measurement.

V. ON SITE EXPERIMENTS AND RESULT ANALYSIS

A. *Experimental platform*

The external structure of the AGV experimental platform used in this article is shown in Figure 8, including battery, power conversion module, power switch, protection frame, load-bearing plate, load, computer, antenna, LiDAR, etc. The internal structure of AGV is shown in Figure 9, which includes a drive module, shock absorber mechanism, PLC, signal processing module, CAN analyzer, etc. In order to improve the utilization of internal space, two 50kg loads are also installed inside the AGV.

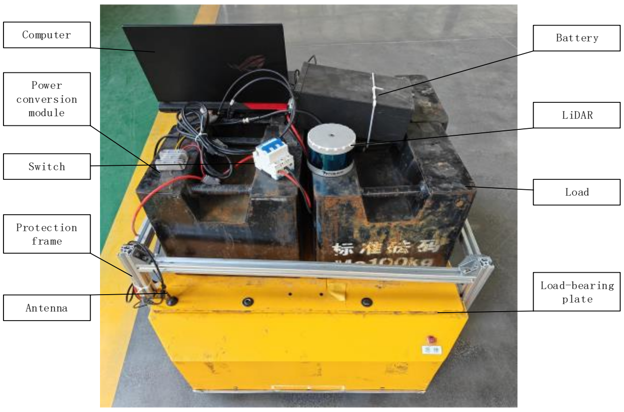

Fig. 8. External structure of AGV

Fig. 9. Internal structure of AGV

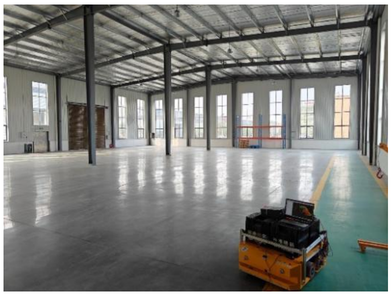

Fig. 10. Experimental Environment

The SLAM experiment in this article was conducted in a workshop of a robot company that produces and tests AGVs, as shown in Figure 10. This experimental site is more in line with the real AGV usage scenario. The experimental route revolves around the workshop from the starting point and then returns to the starting point. The AGV travels straight through the entire process, adjusting its direction in place, and the route is approximately rectangular. By remotely controlling the AGV to move around, the LiDAR can scan the environmental features of the workshop to the maximum extent possible, and finally save the point cloud map in the form of a PCD file to a designated folder.

B. *AGV indoor mapping experiment*

In this experiment, a remote control was used to control the AGV to travel along a closed-loop route in the workshop. To ensure data consistency and subsequent analysis, the ROSBAG related instructions in ROS were used to record the dataset, including data from LiDAR, IMU, and wheel encoder. Figure 11 shows the real-time construction of the workshop 3D map in Rviz using the algorithm proposed in this article. The 3D point cloud map shows clear line driving trajectories. The green part represents the walls, the red part represents the iron gate, window glass, pillars, and the ground, and the blue part represents the machines and other obstacles placed on the ground. The green trajectory line and yellow loop constraint line can be clearly seen on the ground. Figure 12 shows the PCD point cloud map of the workshop, where the red part represents the point cloud composed of line features, and the blue part represents the point cloud composed of face features. Based on the real environment of the workshop, it can be seen that this article effectively created a 3D point cloud map of the real AGV usage environment based on the fusion algorithm of

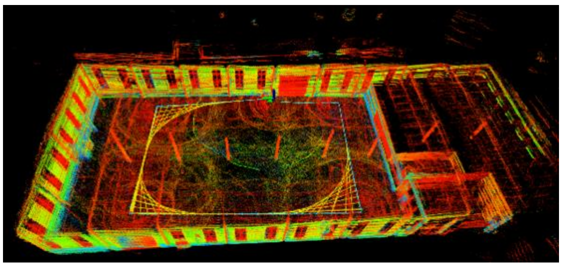

Fig. 11. Three-dimensional diagram of workshop

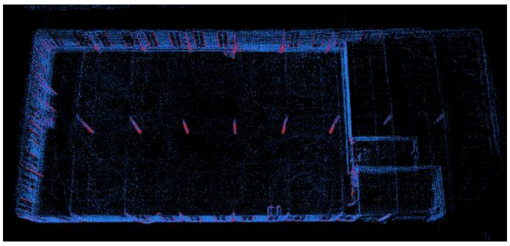

Fig. 12. PCD point cloud map

LiDAR and inertial navigation. The features in the scene were clearly located and restored, and the overall mapping effect did not show drift or ghosting, meeting the requirements for AGV path planning and navigation functions.

C. Slipping experiment

Under normal circumstances, the wheel speed odometry estimates the vehicle's posture changes by monitoring the speed of the AGV wheels. However, when the AGV slips, the speed of the wheels may not match the actual movement speed, resulting in a deviation in the estimated results of the wheel speed odometry. This may cause the wheel speed odometry to mistakenly mistake the tire slip caused by slipping for the actual displacement of the vehicle, resulting in inaccurate pose estimation. When the wheel speed odometry is affected by slipping, the SLAM system may generate inaccurate vehicle pose estimation, which in turn affects the establishment and updating of the map. This may lead to a decrease in the accuracy of the map, and even in severe slip situations, it may cause the SLAM system to lose its ability to accurately track the current position of the vehicle.

By measuring the actual car driving data and comparing it with the point cloud data in ROS., it is obtained that the average pose error of AGV in normal operation is about 0.05 m, and the map accuracy is about 95%.In order to verify the stability and performance of the SLAM system in the event of AGV slippage, the working conditions of AGV slippage were simulated by changing the frictional force between the AGV drive module and the ground.At this point, the left wheel of the AGV experiences continuous slippage while the right wheel drives normally. If the wheel speed odometry is not processed, it will cause significant pose estimation errors and

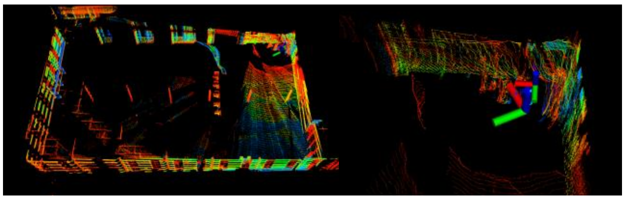

Fig. 13. SLAM without excluding slip

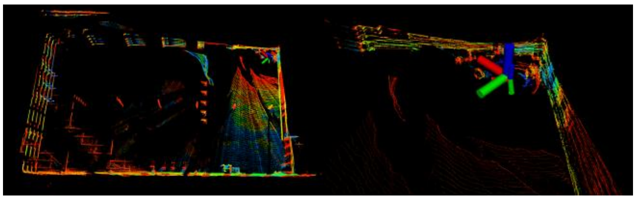

Fig. 14. SLAM Excluding Slipping

mapping errors, The pose error significantly increased to an average of about 0.32 meters, and the map accuracy decreased to about 85%.as shown in Figure 13,At this point, it can be seen that there are obvious errors in the AGV pose estimation, as well as ghosting and drift in the point cloud map. According to the proposed slip state judgment method, the wheel speed odometry during slip is removed, the IMU odometry is directly used, The pose error is reduced to about 0.07m, and the map accuracy is restored to over 95%. The effect is shown in Figure 14, Attitude estimation becomes relatively accurate and the mapping does not produce significant errors.

VI. CONCLUSION

This paper primarily investigates the AGV positioning and mapping algorithm using LiDAR, IMU, and wheel speed odometry as the main sensors. The algorithm employs an Extended Kalman Filter (EKF) to fuse data from the wheel speed odometry and IMU, resulting in high-frequency odometry information with minimal error. The operational status of the AGV is determined based on the slip rate, and the wheel speed odometry is ignored when slippage is detected. In contrast experiments simulating AGV slippage, point cloud maps generated by the algorithm were compared with actual environmental data. The results showed that by discarding erroneous wheel speed odometry during slippage and relying on IMU data, the pose estimation error could be reduced by approximately 78%, and the map accuracy could be restored to over 95%, validating the robustness and accuracy of the algorithm. This study provides a reliable method for AGV positioning and navigation in complex environments, laying a foundation for further improving the stability and accuracy of autonomous driving technology.

REFERENCES

[1] Xu Huan, Liu Qiao, Che Fei, et al. Application of Transport AGV in Material Handling Systems. Logistics Technology, 2013, (06): 80-82.

[2] Yang Wenhua, Su Yunchun, Yu Rui, et al. Application of Contactless Power Supply Technology in AGV. Logistics Technology and Application, 2009, (01): 94-96.

[3] Cao Zhixun, Wang Baohua. Application of Laser Guided Forklifts in China. Logistics Technology and Application, 2006, (7): 98-99.

[4] Che Fei, Liu Qiao, Cui Tieqi. Application of Inertial Navigation AGV in the Electronics Industry. Logistics Technology and Application, 2013, (08): 117-119.

[5] Chu Jiangwei, Gu Baiyuan, Wang Rongben. Design of Two-Wheel Differential Steering AGV Based on Visual Guidance. Logistics Technology, 2002, (07): 12-19.

[6] Wang Rongben. Advantages and Application Prospects of New Vision Navigation AGV. Logistics Technology and Application, 2005, (07): 633-637.

[7] Liu J, Sun Q, Fan Z, et al. TOF lidar development in autonomous vehicle[C]//2018 IEEE 3rd Optoelectronics Global Conference (OGC), Shenzhen, China, 2018: 185-190.

[8] Wang Z, Liao J, Guo H, et al. Lidar/IMU Integrated Navigation and Positioning Method[C]//International Conference on Multimedia Technology and Enhanced Learning.Springer, Cham, online, 2021: 407-413.

[9] Ji L, Zhao L, Duo J, et al. A Novel 3D Lidar-IMU Calibration Method Based on Hand-Eye Calibration System[C]//Chinese Intelligent Systems Conference. Springer, Singapore, Jiujiang, China, 2020: 342-350.

[10] Gao Yunfeng, Zhou Lun, Lv Mingrui, Liu Wentao. A Review of Indoor Positioning Methods for Autonomous Mobile Robots. Sensors and Microsystems, 2013, 32(12): 1-5+9.